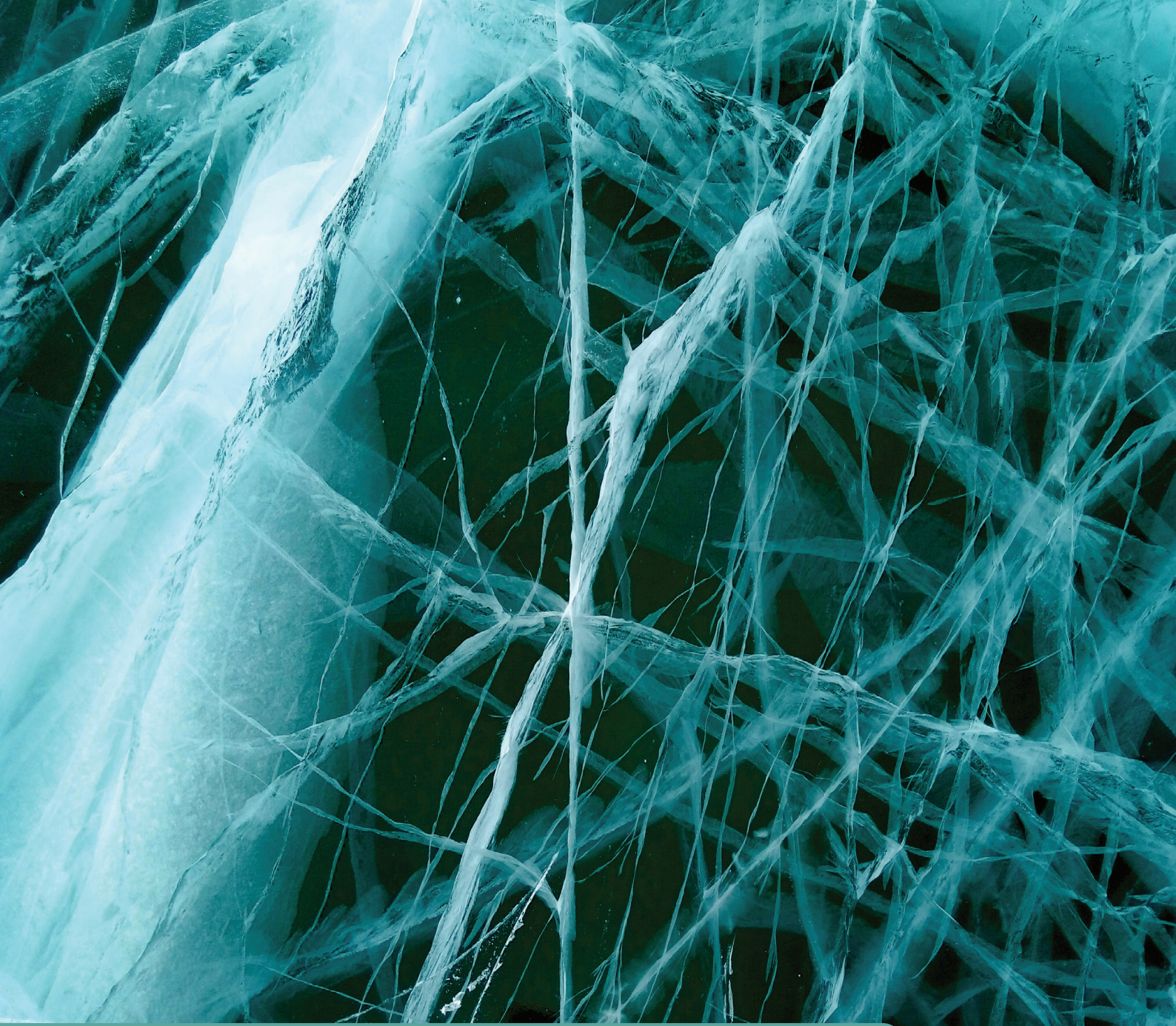


PDQeX *phyto*GEM

PDQeX *phyto*GEM System
Rapid, automated DNA extraction from plants

Exymes

BR010 V1

The PDQeX

The PDQeX is a compact, programmable temperature control unit that automates enzymatic nucleic acid extraction.

It steps each sample through the incubation temperatures required by the *phyto*GEM chemistry, then delivers the extract into a collection tube without centrifugation, spin columns or manual transfer steps.

Each extraction takes place in a sealed tube, which reduces handling and the risk of cross-contamination between samples.

Its small footprint and low weight make the instrument portable, so it can be used at the bench or moved to where the samples are collected.



The Disposable Extraction Cartridge

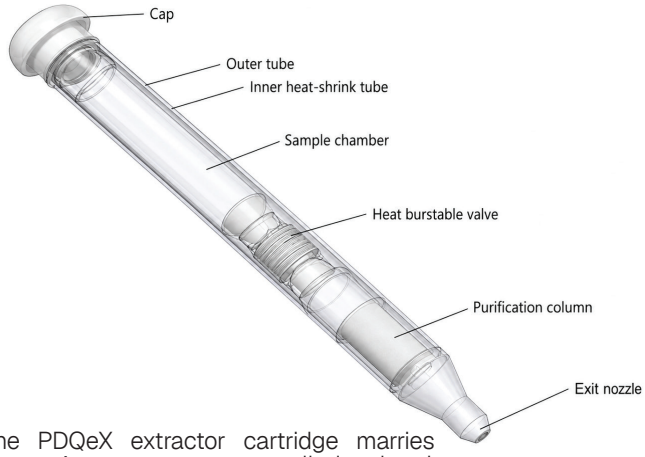
Extracting nucleic acids from biological samples can be laborious and frustrating. The PDQeX system from Exymes takes the tedium away by combining powerful enzyme-driven extraction chemistries with an innovative extraction cartridge.

DNA extraction using the PDQeX technology, is rapid and automated, going from leaf on a plant to PCR-ready DNA in approximately 15 minutes.

The chemistry uses potent thermophilic proteinases combined with a cocktail of mesophilic cell wall degrading hydrolases. These enzymes systematically lyse cells, destroy nucleases, digest the protein and release the DNA.

Extractions are performed in disposable cartridges that are designed to remove inhibitory polyphenols and polysaccharides.

The *phyto*GEM system can be easily adapted to work with different types of plant tissue and different samples.



The PDQeX extractor cartridge marries Exymes' temperature-controlled chemistry with thermo-responsive polymers that pump the DNA through a purification column

What sets *phyto*GEM apart?

- Easy sampling and long term storage solution
- No centrifugation
- Single-step extraction and purification
- Superior inhibitor removal
- Sample to PCR-ready DNA in 20 mins
- Closed-tube system prevents contamination.

The PDQeX phytoGEM Sampling System

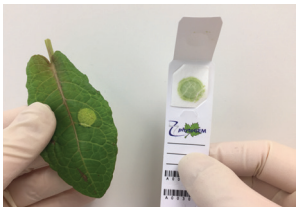
Sampling takes just seconds. Leaves are crushed in the field onto a storage card using a special tool. With long-term storage of samples built into the procedure, there is no extra work or freezers packed with bags and tubes.



The jaws of the crusher tool can lock for easy packing. To unlock them, squeeze the handles fully and release. The jaws will then open.



Remove a PhytoCard from its sterile pack and lift the flap covering the absorbant card. Slide the card fold-first into the groove of the jaws.

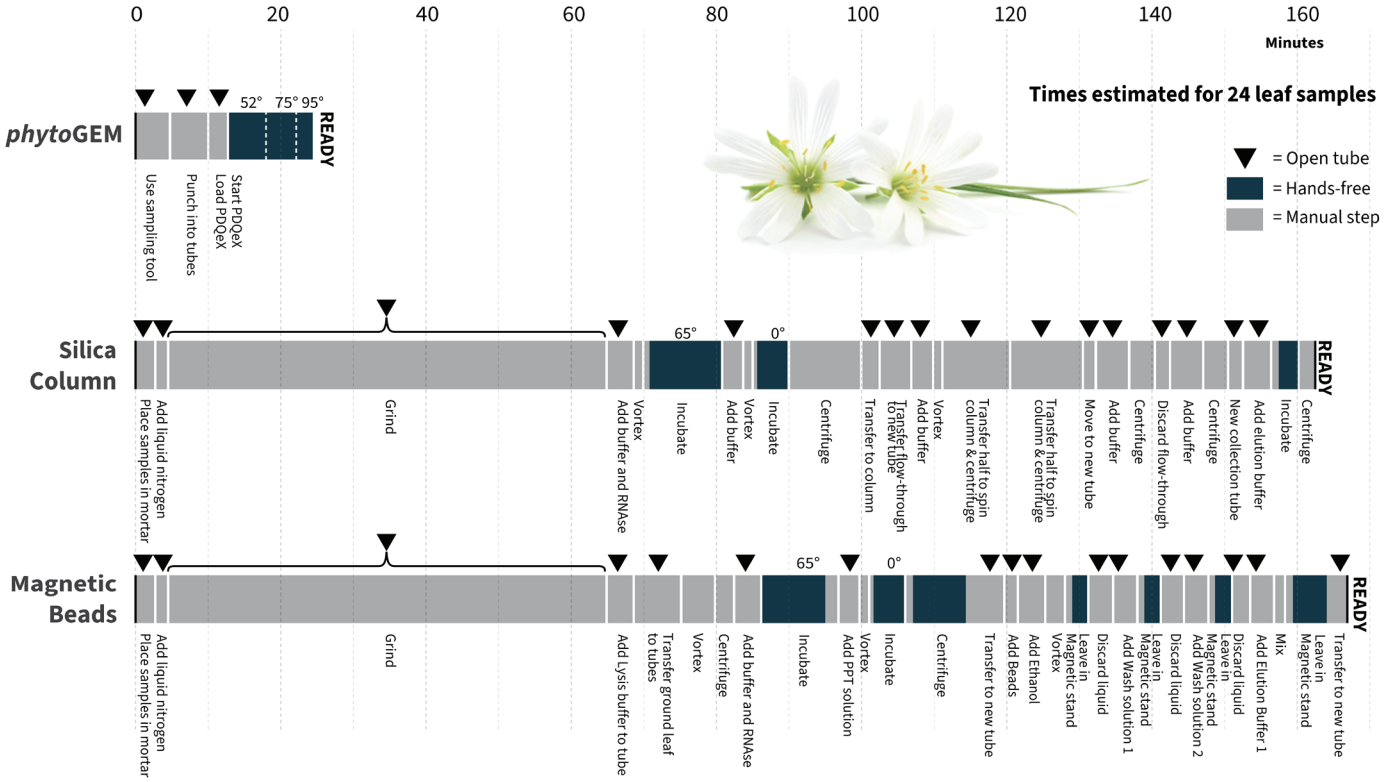


Place the area of the leaf that you wish to sample over the absorbent card and close the jaws firmly until they release.



And that's all. Simple! Your leaf sample will be crushed onto the card and is now ready for extraction or storage. . Sampling, storage and DNA extraction in a few easy steps

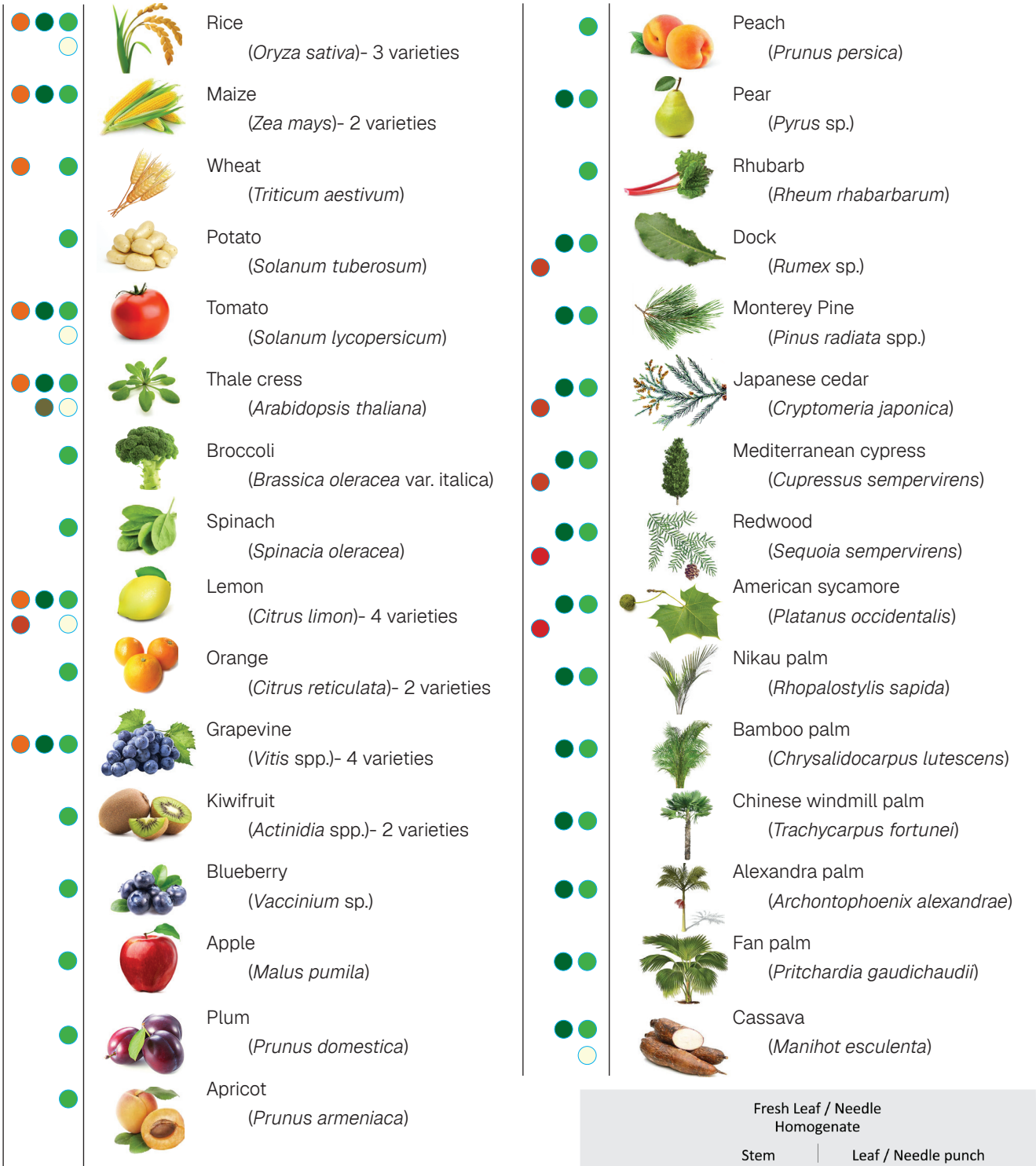
The difference...

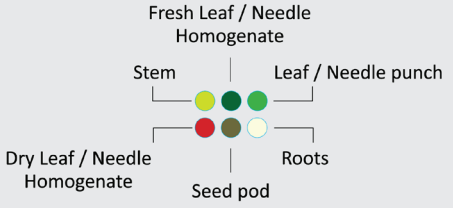


Validated plant species and tissues using the PDQeX *phyto*GEM system

Each plant species has its own unique difficulties for nucleic acid extraction. This can be due to a number of causes: sample toughness, different inhibitors, polyphenols or different polysaccharides. The *phyto*GEM system has been tested and optimised on a wide variety of species and tissue types. The table below shows where we are so far. This list is by no means comprehensive but is growing week by week.

If you have your own favourite plant and it is not on this list, contact us from our website. We should be able to help you.





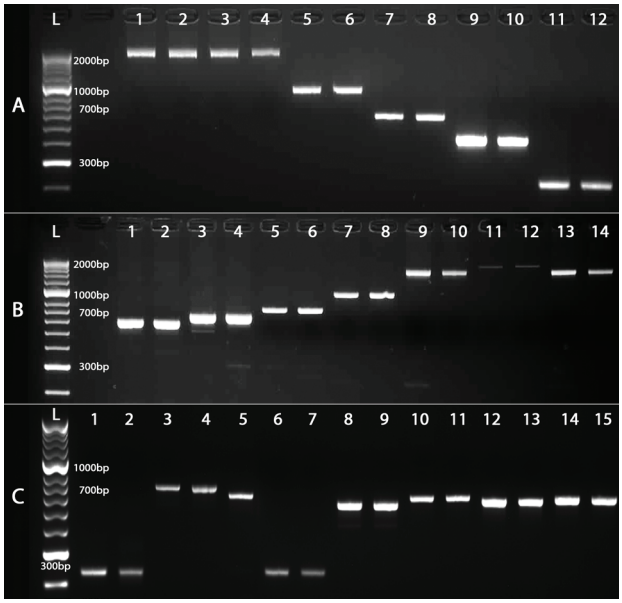
Plants and plant tissues in the *phyto*GEM repertoire (so far).

Typical results using *phyto*GEM

DNA yields will vary from species to species. Some example numbers are in the table below. These were obtained from three x 3 mm punches from the storage card.

Plant	Yield (ng in 100 µl)
Arabidopsis thaliana	150 – 218
Oryza sativa	103 – 299
Citrus limon	96 – 166
Triticum aestivum	110 – 132
Solanum lycopersicum	282 – 360
Zea mays	90 – 152
Solanum tuberosum	96 – 154
Vitis spp	144 – 256
Prunus domestica	80 – 278
Malus pumila	241 – 347

The DNA is suitable for standard end point PCR....

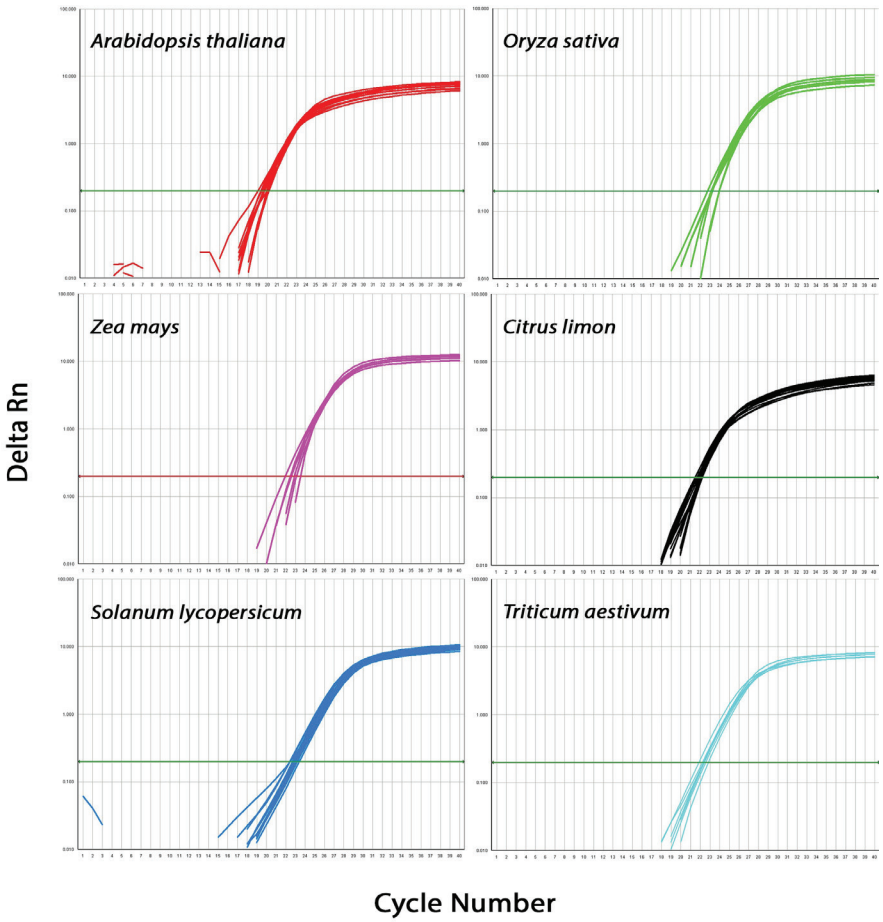


Agarose gel images showing amplicons generated from plant extractions.
Fragments were generated using primer combinations targeting sequences in 5 genes: *adh* (alcohol dehydrogenase), *act*2 (actin 2), *mat*k (maturase K), *rbc*L (RuBisCO large subunit) and *psb*A (photosystem II) ranging from ~200–2300 bp.

(A) *Arabidopis:* 1-4: *act*2, 5-6: *mat*K, 7-8: *rbc*L, 9-10 *psb*A, 11-12: *adh*
(B) Rice: 1-2: *rbc*L, 3-4: *psb*, 5-14: various fragments of *adh*.
(C) 1-2: lemon *adh*, 3-4: lemon *rbc*L, 5: lemon *psb*, 6-7: maize *adh*, 8-9: maize *psb*, 10-11: maize *rbc*L, 12-13: tomato *psb*, 14-15: tomato *rbc*L.

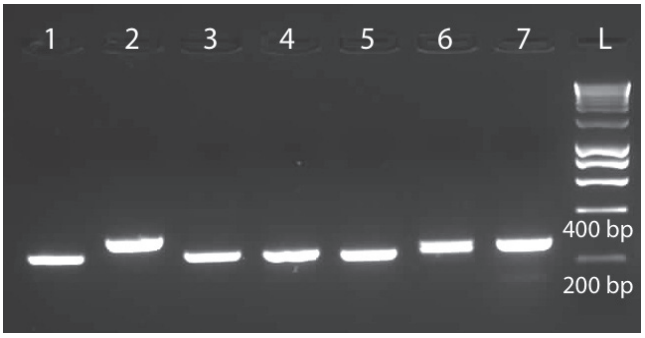
Typical results using *phyto*GEM

... and is also of excellent quality for qPCR



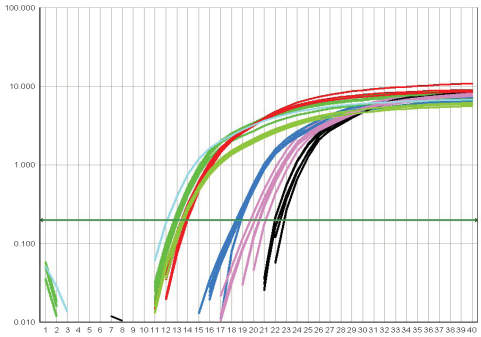
qPCR plots of alcohol dehydrogenase (adh) gene fragments amplified from phytoGEM extractions from 6 plants.

The *phyto*GEM system can also be used for extracting fungal DNA



Agarose gel separation ITS amplicons from:

1. Cladosporium sp.;
2. Cadophora malorum;
3. Geomyces sp.;
4. Aspergillus sp.;
5. Penicillium sp.1;
6. Penicillium sp. 2;
7. Penicillium sp. 2.



qPCR plots of amplified ITS regions.

Cladosporium sp.;
Cadophora malorum;
Geomyces sp.;
Aspergillus sp. 2;
Penicillium sp. 1 3;
Penicillium sp. 2;
Penicillium sp. 2

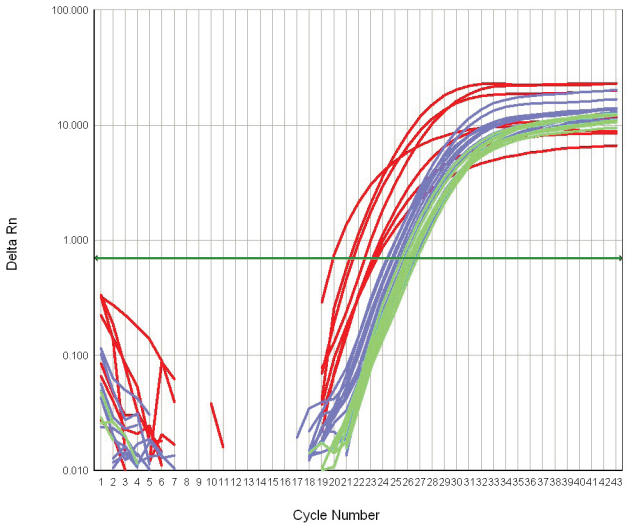
*phyto*GEM for detection of plant pathogens

Early detection of pathogens is vital for successful crop management. It allows proactive control measures to be applied well before the establishment and spread of a disease. The phytoGEM system offers a simple hands-off method for DNA extraction and is ideal for studying endophytic plant pathogens.

So far, *phyto*GEM has been validated for the detection of *Pseudomonas syrigae pv. actinidiae* (Psa) detection in kiwifruit, *Phytophthora spp.* in kauri, pine oak, *Phytoplasma spp.* in grapevines, and Xyllela fastidiosa in a variety of plants.



A kiwifruit leaf infected with *Pseudomonas syrigae pv. actinidiae.*



qPCR plots showing the presence of Psa-V DNA in leaf infected under controlled conditions.
Leaf samples directly on infected brown spots;
Leaf samples adjacent to spots;
Plots from crushes made on clear area of infected leaf.

*phyto*GEM for RNA?

This research is nearly complete and is due for release in 2027.

ORDERING / DISTRIBUTION

Description	Product code
PDQeX Machine	XGE0001
PDQeX Plant Starter Kit (includes crusher tool, punches and phytoCards)	PPS0001

Description	Quantity	Product code
phytoGEM Cartridges (*phyto*GEM extraction kit and PDQeX tubes)	100 500 1000	XPP0100 XPP0500 XPP1000
phytoCards	500 1000	PCA0500 PCA1000
Sampling tool	1	PSC001

Distribution

Information about Exymes' s distributors can be found on the Exymes Website: www.exymesplc.com

info@exymesplc.com | www.exymesplc.com

At Exymes, our goal is to provide ultra fast nucleic extraction enabling a broader spectrum of users to both employ and benefit from molecular techniques. The first step is the simplification of sample preparation. Our temperature-driven, single-tube process simplifies and reduces the number of steps for traditional nucleic acid extraction, resulting in high-quality extracts with reduced contamination and high yields - all in minutes, not hours.

Committed to minimal packaging, reduced plastic use and a sustainable world

Exymes

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